higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

INDUSTRIAL INSTRUMENTS N5

(8080205)

18 April 2024 (X-paper)
09:00–12:00

Drawing instruments and nonprogrammable calculators may be used.

This question paper consists of 6 pages and 1 formula sheet.

301Q1A2418

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
INDUSTRIAL INSTRUMENTS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Use only black or blue ink.
5. Write neatly and legibly.

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SECTION A: FLOW MEASUREMENT

QUESTION 1

1.1 Name the TWO methods that must be incorporated into secondary devices to compensate for the square root problem. (2)

1.2 Name THREE types of integrators. (3)

1.3 Make a neat, labelled sketch of an ultrasonic flow meter. (5)

1.4 State THREE advantages and TWO disadvantages of an orifice plate. (5)

[15]

QUESTION 2

2.1 Explain what is meant by the *square root problem*, with reference to differential pressure and flow. (4)

2.2 Explain the working principle of a variable area flowmeter. (6)

2.3 Where do we take the reading with the meter mentioned in QUESTION 2.2? (2)

2.4 What is an integral orifice differential pressure cell, and where can it be used? (3)

2.5 Choose a term from COLUMN B that description in COLUMN A. Write only the letter (A–F) next to the question number (2.5.1–2.5.6) in the ANSWER BOOK.

COLUMN A		COLUMN B	
2.5.1	The purpose of this instrument is to generate a differential of the flow.	A	feedback loop
2.5.2	The instrument that measures the total volume of flow in a certain time.	B	square root problem
2.5.3	The system will send a signal to the process flow control valve to be corrected.	C	primary instrument
2.5.4	This controls the process for an output for an instrument and if an error occurs to the feedback system.	D	control valve
2.5.5	The flow rate is the square root of the differential pressure.	E	integrator
2.5.6	The instrument that controls the amount of flow of substance.	F	process control

(6 × 1) (6)

[21]

TOTAL SECTION A: 36

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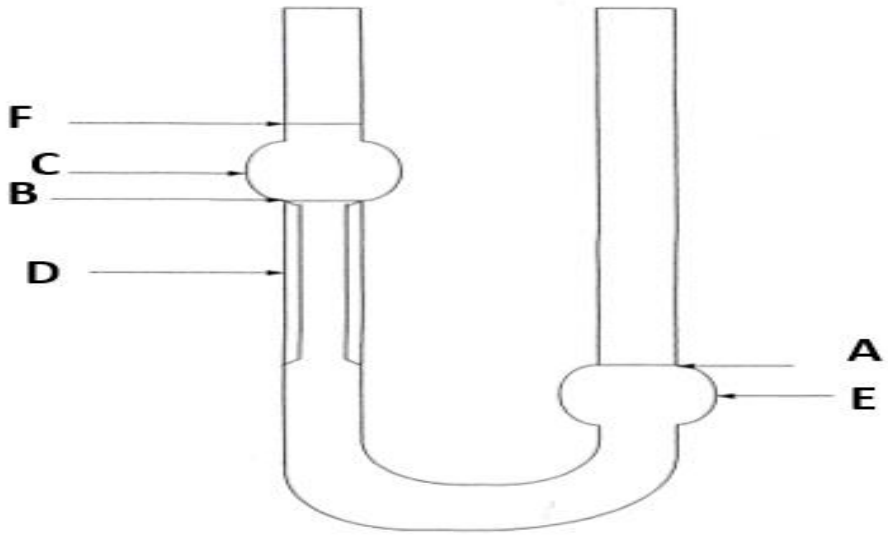
SECTION B

QUESTION 3: DENSITY, HUMIDITY & VISCOSITY MEASUREMENT

3.1 Make a neat, labelled sketch of a weighing tube hydrometer for continuous measurement of fluid density. Your sketch must clearly indicate how a pneumatic output is generated. (5)

3.2 How would you compensate for temperature changes when using the hydrometer mentioned in QUESTION 3.1? (2)

3.3 Study the figure below and answer the following questions:



3.3.1 Name the type of the viscometer shown by the sketch. (1)

3.3.2 Label the parts labelled A to F. (6)

3.4 Name THREE different types of a float viscometer. (3)

3.5 Sketch a piezo-electric sampling system. Your sketch must clearly indicate how the measuring and reference gas is switched. (8)

[25]

QUESTION 4: pH AND CONDUCTIVITY MEASUREMENT

4.1 It is possible to measure pH by colour change of certain pH dye, by using certain metal oxide electrodes, and by various other techniques.

Name the THREE most commonly used pH measuring methods. (3)

4.2 All materials likely to come into contact with the process liquid must be chemically resistant at all measurement temperatures.

Name THREE properties that the assembly and sealing materials must have for the ideal housing of materials that come into contact with the process liquid. (3)

4.3 The electrodes in their simple state may not be suitable for industrial processes without further protection.

Name the TWO methods to overcome this disadvantage. (2)

4.4 Metallic electrodes are subjected to poisoning by trace quantities of impurities which are found in electrolytic solutions, and in some cases by exposure to the atmosphere. Therefore, precautions must be taken in preparing and storing electrodes in order to obtain reliable readings on the various systems.

Name FOUR precautions to be taken in preparing and storing electrodes. (4)

4.5 Conductivity measurement systems that utilise simple ohm meters suffer from inaccuracy due to line voltage variations.

4.5.1 How would you overcome the disadvantage?

4.5.2 How would you increase sensitivity and reduce non-linearity caused by meter resistance?

(2 × 2) (4)
[16]

QUESTION 5: AUTO CONTROL

5.1 Name any FOUR types of delays that happens in process control. (4)

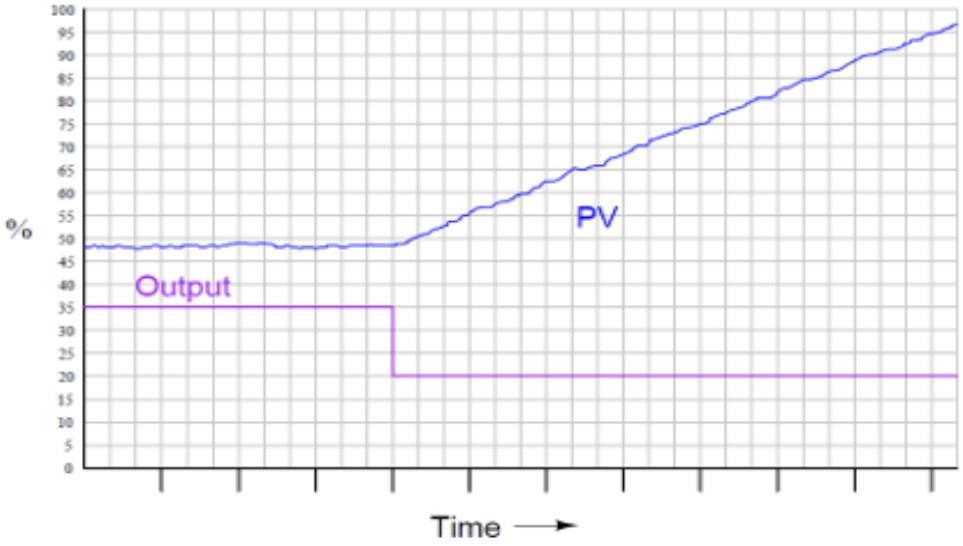
5.2 When we want to control a process we must be able to discriminate between the set point and process variable. Two methods of discrimination are most commonly used, namely position balance and force balance.

5.2.1 Make a neat, labelled sketch of the force balance discrimination method. (6)

5.2.2 Explain what happens to this system when there is no error. (2)

5.2.3 Explain what happens to this system when there is an error and how that error can be rectified. (3)

5.3 The open-loop response of the process is shown in the following trend recording. What type of control is indicated by the diagram below?



(2)

5.4 Make a neat, labelled sketch of curves, showing the effect of a narrowing proportional band on recovery curves following set point load change. The curves must show how proportional band adjustment is a compromise between instability and offset error. (6)

[23]

TOTAL SECTION B: 64
GRAND TOTAL: 100

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INDUSTRIAL INSTRUMENTS N5

FORMULA SHEET

$$Q = \frac{8}{15} \quad Tan \frac{\theta}{2} \sqrt{2g.H^5}$$

$$Q = \frac{2}{3} \; B\,C\sqrt{2g.H^3}$$

$$Q = \frac{2}{3} \; CB\sqrt{2g.H^3}$$

pH = -log[H^+]

pOH = -log [OH^-]

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